## Restriction endonucleases: Molecular scissors

The restriction enzymes are called as molecular scissors. These act as foundation of recombinant DNA technology. These enzymes exist in many bacteria where they function as a part of their defence mechanism called restriction-modification system.

There are three main classes of restriction endonucleases : Type I, Type II and Type III, which differ slightly by their mode of action. Only type II enzyme is preferred for use in recombinant DNA technology as they recognise and cut DNA within a specific sequence typically consisting of 4-8 bp. Examples of certain enzymes are given in table 4.1.

The restriction enzyme ***Hind II*** always cut DNA molecules at a point of recognising a specific sequence of six base pairs. This sequence is known as recognition sequence. Today more than 900 restriction enzymes have been isolated from over 230 strains of bacteria with different recognition sequences.

This sequence is referred to as a restriction site and is generally –palindromic which means that the sequence in both DNA strands at this site read same in 5' – 3' direction and in the 3'-5' direction

Example: MALAYALAM: This phrase is read the same in either of the directions.

Restriction endonucleases are named by a standard procedure. The first letter of the enzymes indicates the genus name, followed by the first two letters of the species, then comes the strain of the organism and finally a roman numeral indicating the order of discovery. For example, ***EcoRI*** is from Escherichia (**E**) coli (**co**), strain RY 13 (**R**) and first endonuclease (**I**) to be discovered.

**Palindromic repeats**: A symmetrical repeated sequence in DNA strands

5' ... CATTATATAATG ... 3'

3' ... GTAATATATTAC ... 5'

Note: That the sequence of the base pairs in the reverse direction when compare to the first sequence.

The exact kind of cleavage produced by a restriction enzyme is important in the design of a gene cloning experiment. Some cleave both strands of DNA through the centre resulting in **blunt** or **flush end**. These are known as symmetric cuts. Some enzymes cut in a way producing protruding and recessed ends known as **sticky** or **cohesive end**. Such cut are called staggered or asymmetric cuts.

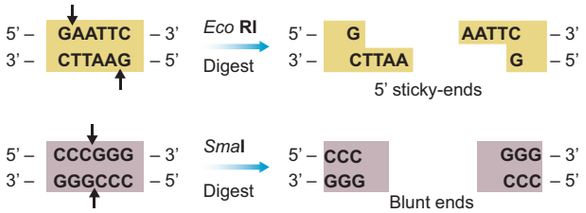



**Figure 4.5:** Sticky and Blunt ends

Two other enzymes that play an important role in recombinant DNA technology are DNA ligase and alkaline phosphatase

### 4.4.2 DNA Ligase

DNA ligase enzyme joins the sugar and phosphate molecules of double stranded DNA (dsDNA) with 5'-$PO_4$ and a 3'-OH in an Adenosine Triphosphate (ATP) dependent reaction. This is isolated from T4 phage.

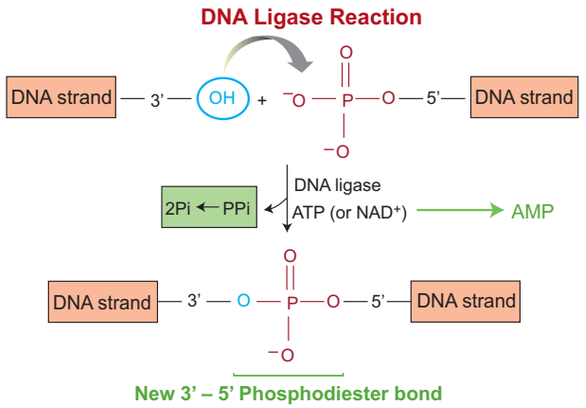



**Figure 4.6:** DNA ligase reaction

### 4.4.3 Alkaline Phosphatase

It is a DNA modifying enzyme and adds or removes specific phosphate group at 5' terminus of double stranded DNA (dsDNA) or single stranded DNA (ssDNA) or RNA. Thus it prevents self ligation. This enzyme is purified from bacteria and calf intestine.



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